Domain Knowledge Document

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1 Domain Knowledge Document

2 Grocery Inventory & Sales Performance Dashboard

2.1 Microsoft Excel Data Analytics Project

Project Name: Grocery Inventory & Sales Performance Dashboard

Domain: Grocery Retail & Inventory Management

Tool Used: Microsoft Excel

Project Type: Data Analytics Portfolio Project

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4 1. Introduction

The grocery retail industry is one of the fastest-moving sectors, where businesses manage hundreds of products across multiple categories every day. Efficient inventory management and sales monitoring are essential to ensure product availability, maximize revenue, and reduce inventory costs.

This project focuses on developing an interactive Microsoft Excel dashboard that transforms raw inventory and sales data into meaningful business insights. The dashboard provides management with real-time visibility into sales performance, inventory value, monthly revenue trends, and product replenishment requirements.

Understanding the grocery retail domain is essential for selecting appropriate KPIs, creating meaningful visualizations, and supporting business decision-making.

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5 2. Grocery Retail Industry Overview

A grocery retail business purchases products from suppliers and sells them to customers through physical stores or online channels.

The success of a grocery business depends on maintaining the right balance between product availability and inventory investment.

The major operational goals include:

- Maintaining sufficient stock levels.
- Reducing stock shortages.
- Avoiding excess inventory.
- Improving inventory turnover.
- Increasing sales revenue.
- Meeting customer demand.
- Supporting profitable business operations.

The dashboard developed in this project helps management monitor these objectives through interactive reporting.

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6 3. Inventory Management

Inventory Management is the process of tracking, controlling, and maintaining stock levels to ensure products are available whenever customers require them.

Proper inventory management helps businesses:

- Prevent stock shortages.
- Reduce overstocking.
- Lower inventory carrying costs.
- Improve inventory utilization.
- Increase customer satisfaction.

- Support efficient replenishment planning.

6.0.1 Dataset Fields Used

- Product_ID
- Product_Name
- Stock_Quantity
- Reorder_Level
- Reorder_Quantity

6.0.2 Dashboard Components

- Total Products
- Inventory Value
- Reorder Products
- Reorder Percentage
- Reorder Status Distribution

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7 4. Sales Performance Management

Sales Performance measures how effectively products generate revenue over time.

Monitoring sales performance helps management:

- Identify high-performing products.
- Compare category-wise sales.
- Evaluate monthly business performance.
- Support marketing and promotional decisions.

7.0.1 Dataset Fields Used

- Sales_Volume
- Unit_Price

7.0.2 Dashboard Components

- Total Revenue
- Total Units Sold
- Revenue by Category
- Monthly Revenue Trend
- Top 10 Products by Revenue

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8 5. Revenue Analysis

Revenue represents the total income generated from product sales.

8.0.1 Business Formula

`Revenue = Sales Volume × Unit Price`

Revenue analysis helps businesses:

- Measure sales performance.
- Identify top-performing categories.
- Compare product performance.
- Monitor business growth.

8.0.2 Dashboard Components

- Total Revenue KPI
- Revenue by Category
- Monthly Revenue Trend
- Top 10 Products by Revenue

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9 6. Inventory Value Analysis

Inventory Value represents the total monetary value of products currently available in stock.

9.0.1 Business Formula

`Inventory Value = Stock Quantity × Unit Price`

Monitoring inventory value helps businesses:

- Understand inventory investment.
- Optimize stock allocation.
- Reduce carrying costs.
- Improve inventory planning.

9.0.2 Dashboard Components

- Inventory Value KPI
- Inventory Value by Category

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10 7. Reorder Management

Reorder Management ensures products are replenished before stock levels become critically low.

Each product has a predefined **Reorder Level**.

10.0.1 Business Rule

`If`

```
Stock Quantity < Reorder Level

↓

Reorder Required

Otherwise

↓

Sufficient Stock
```

The dashboard automatically identifies products requiring replenishment.

10.0.2 Dashboard Components

- Reorder Products KPI
- Reorder Percentage KPI
- Reorder Status Distribution
- Reorder Status Filter

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11 8. Inventory Turnover Rate

Inventory Turnover Rate measures how efficiently inventory is sold and replenished during a given period.

A higher turnover rate indicates:

- Faster product movement.
- Efficient inventory management.
- Lower inventory holding costs.

A lower turnover rate may indicate:

- Overstocking.
- Slow-moving inventory.
- Reduced customer demand.

11.0.1 Dashboard Component

Average Inventory Turnover Rate by Category

This visualization enables management to compare inventory efficiency across different grocery categories.

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12 9. Monthly Revenue Trend Analysis

One of the major enhancements in the final dashboard is the **Monthly Revenue Trend** visualization.

This line chart displays monthly revenue generated throughout the selected year.

12.0.1 Purpose

- Monitor monthly business performance.
- Identify seasonal demand.
- Compare monthly revenue.
- Evaluate business growth.
- Support inventory planning.

12.0.2 Dataset Fields Used

- Cleaned_Date_Received
- Revenue

12.0.3 Dashboard Visualization

Chart Type: Line Chart

12.0.4 Business Value

The Monthly Revenue Trend enables management to:

- Detect seasonal sales patterns.
- Compare revenue across months.
- Identify peak and low-performing months.
- Make better inventory and sales planning decisions.

The Year Timeline filter allows users to switch between **2024** and **2025**, making the analysis interactive.

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13 10. Key Performance Indicators (KPIs)

The dashboard summarizes business performance using the following KPIs.

KPI	Business Purpose
Total Products	Total inventory available
Total Revenue	Overall sales performance
Total Units Sold	Customer demand
Inventory Value	Current inventory investment
Reorder Products	Products requiring replenishment
Reorder Percentage	Inventory health indicator
Average Inventory Turnover Rate	Inventory efficiency

These KPIs provide a quick overview of business performance.

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14 11. Business Metrics

The dashboard uses the following business metrics.

Metric	Description
Revenue	Sales Volume $\times$ Unit Price
Inventory Value	Stock Quantity $\times$ Unit Price
Total Units Sold	Sum of Sales Volume
Reorder Products	Count of products requiring reorder
Reorder Percentage	Percentage of products requiring replenishment
Inventory Turnover Rate	Average inventory turnover
Monthly Revenue	Sum of monthly revenue

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15 12. Dashboard Visualizations

The dashboard contains the following visualizations.

15.1 KPI Cards

- Total Products
- Total Revenue
- Total Units Sold
- Inventory Value
- Reorder Products
- Reorder Percentage
- Average Inventory Turnover Rate

15.2 Charts

- Monthly Revenue Trend
- Revenue by Category
- Inventory Value by Category
- Top 10 Products by Revenue
- Inventory Turnover Rate by Category
- Reorder Status Distribution

15.3 Interactive Filters

- Category
- Year Timeline
- Reorder Status

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16 13. Business Questions Answered

The dashboard answers several important business questions.

16.0.1 Sales Performance

- What is the total revenue generated?
- Which category generates the highest revenue?
- Which products generate the highest revenue?
- How does revenue change month by month?

16.0.2 Inventory Performance

- What is the current inventory value?
- Which category holds the highest inventory value?
- How efficiently is inventory moving?

16.0.3 Replenishment

- Which products require replenishment?
- What percentage of products require reorder?

16.0.4 Business Performance

- Which categories perform best?
- How does performance vary across different years?

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17 14. Business Benefits

The dashboard provides several business benefits.

- Centralized reporting.
- Improved inventory visibility.
- Better sales monitoring.
- Faster decision-making.
- Efficient inventory planning.
- Improved replenishment management.
- Monthly revenue monitoring.
- Reduced manual reporting effort.
- Interactive business analysis.

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18 15. Key Business Terminologies

Term	Meaning
Inventory	Products available for sale

Term	Meaning
Stock Quantity	Current quantity available
Reorder Level	Minimum stock before replenishment
Reorder Quantity	Quantity to be ordered
Revenue	Income generated from sales
Inventory Value	Monetary value of inventory
Sales Volume	Total units sold
Inventory Turnover Rate	Inventory movement efficiency
KPI	Key Performance Indicator
Dashboard	Interactive business reporting solution

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19 16. Conclusion

Understanding the grocery retail domain is essential for developing meaningful dashboards and supporting business decisions.

The **Grocery Inventory & Sales Performance Dashboard** combines domain knowledge with Microsoft Excel's analytical capabilities to transform raw inventory and sales data into actionable business insights.

Through KPI cards, monthly revenue trend analysis, inventory monitoring, category performance evaluation, and interactive filtering, the dashboard enables management to monitor business performance efficiently and make informed operational decisions.

This project demonstrates practical knowledge of **inventory management, sales analysis, business reporting, KPI development, dashboard design, and data visualization**, making it a valuable portfolio project for aspiring Data Analysts.